Yash Maini

B.Tech AI & ML, GGSIPU '26 | GATE '25 Top 8% | 2× IEEE First Author
New Delhi, India | +91-8920028757 | mainiyash2@gmail.com | yashmaini.tech
linkedin.com/in/yashmaini | github.com/Yashmaini30 | leetcode.com/yashmaini

Summary

Engineer (B.Tech '26) building **production LLM agent systems and the Python backends behind them** – async voice pipelines at **sub-600ms**, a multi-agent orchestration SDK unifying **8 agents**, and an **11-tool LangGraph Salesforce copilot** spanning frontend, CRM writes and NL-to-SQL analytics. Curated a **1.5B-token** CoT fine-tuning corpus; 2× IEEE first author; MLOps across AWS and GCP.

Experience

NetoAI Solutions — **Jan 2026 – Present**
Generative AI Engineer Intern — *Remote*

- Replaced manual PBX workflows for enterprise client **Siemcom** by architecting a **production LLM voice agent** on FreeSWITCH + SIP, orchestrating real-time **STT → LLM → TTS** across concurrent multi-channel calls and owning integration through go-live.
- Cut end-to-end voice latency **40%** to under **600ms** under concurrent load by re-engineering the **async Python inference layer** with asyncio, streaming generation and parallelized TTS.
- Cut new-agent integration time **60%** by building a **multi-agent orchestration SDK** that unified **8 agents** behind one interface with standardized tool-calling, schema-validated I/O and intent-based routing.
- Automated a **4-stage** B2B sales pipeline across **7 Salesforce object types** by building an **11-tool LangGraph copilot** wiring **frontend → backend → agent → Salesforce REST API**, with **4-vendor** LLM routing (Mistral/OpenAI/Google/vLLM), branded PDF proposal generation and Google Meet scheduling.
- Gave non-technical reps self-serve access to live CRM data via **natural-language SOQL analytics** behind a read-only query-safety validator and **2 explainable scoring models** (deal-health, BANT lead scoring; 0–100 → Hot/Warm/Cold); hardened with **OAuth** token reuse/retry, Postgres-backed memory failover and **32 unit tests** on a Dockerized gunicorn deploy.
- Reduced production hallucination rates by curating a **1.5B-token** domain fine-tuning corpus with structured **chain-of-thought** annotations and building **RAG** pipelines with semantic search over it.

AIIMS Delhi – Medical Imaging Research — **Jul 2025 – Sep 2025**
Research Intern – Automated Pipeline Deployment — *New Delhi*

- Designed and deployed an **end-to-end automated ingestion pipeline** (Python, PostgreSQL, Docker, Orthanc PACS) achieving **95% workflow automation** – cutting processing from hours to **30 seconds per video**.
- Delivered under **Dr. Deepak Agrawal** (Prof. of Neurosurgery, AIIMS Delhi); awarded a **Letter of Recommendation** for production-quality engineering and clinical data integrity.

Education

University School of Automation and Robotics, GGSIPU — **Nov 2022 – Jun 2026**
B.Tech – Artificial Intelligence and Machine Learning | CGPA: 8.61/10 — *New Delhi, India*

Technical Skills

GenAI & LLMs: Fine-tuning (LoRA/QLoRA, PEFT), RAG, AI Agents, Multi-Agent Orchestration, Tool Calling, Evals, LangChain/LangGraph, vLLM
Backend & APIs: Python, FastAPI, Flask, AsyncIO, REST, WebSockets, Microservices, Salesforce/SMTP integrations, Bash
Data & ML: pandas, NumPy, scikit-learn, PyTorch, TensorFlow, PostgreSQL (pgvector), MongoDB, SQL optimization, Embeddings, FAISS/Pinecone
MLOps & Cloud: Docker, CI/CD, MLflow, Terraform, AWS (Bedrock, SageMaker, Lambda, EC2, S3), GCP (Vertex AI, BigQuery)

Projects

CopyGuard – Serverless GenAI Code Detection Platform | GitHub | Demo
- **AWS Lambda, Bedrock Claude, Terraform, S3, CloudFront, Grafana** – serverless GenAI detection service handling burst traffic at **under 2s response** and **99.9% uptime**; entire infra as code, zero manual ops.

ThreatMatrix – MLOps Pipeline for Network Intrusion Detection | GitHub | Demo
- **Python, FastAPI, MongoDB, MLflow, Docker, AWS EC2** – modular ingest→validate→train→serve pipeline with full CI/CD; FastAPI inference endpoint at **sub-15ms latency** with MLflow tracking and model registry.

Candida Hyphae Segmentation – Deep Learning for Medical Imaging
- **PyTorch, EfficientNet-B4, A100 GPU** – custom Gabor-guided UNet for clinical image segmentation achieving **Dice 0.72 | IoU 0.56 | Recall 0.94** across **86,000 tiles**; **38× faster** data pipeline.

Publications

- **Y. Maini** (First Author) et al., "Breast Tumor Classification with Fine-Tuned Hyperparameter Training using Deep Learning Models," **2025 AI-Driven Smart Healthcare**, Kolkata – IEEE DOI
- **Y. Maini** (First Author) et al., "Xception for Tomato Leaf Disease Detection: Hyperparameter Tuning and Fine-tuning," **2024 ICAIQSA**, Nagpur – IEEE DOI

Certifications & Achievements

Credentials: OCI Generative AI Professional, OCI Data Science Professional, **IBM AI Agent Architect**
Achievements: GATE 2025 Top 8% (DS&AI); **2× IEEE First Author**; **Top 12% LeetCode** (700+ problems); Coding Society Coordinator – 2 university-wide events, 200+ teams